Case Study Interview for Data Science and MLOps Master Thesis



A message for the candidate

Dear candidate,

We are pleased to let you know that you have passed the initial application screening process. We would like to invite you to the next stage of internship application with a programming and data science case study.

Your case study results will be evaluated based on the tasks completion as well as clear reasoning written in both code commenting and Sphinx documentation. You will be invited to an interview if you pass the case study evaluation. All the best in your case study!

We are looking forward to receiving your results.

Data Science Hiring Committee

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Case study questions

In this case study, you will find a sample dataset saved in the parquet format. The dataset is contaminated with different forms of anomalies. Your tasks:

- Analyze and interpret the measurement data to identify trends and patterns.
- Improve data quality with statistics and machine-learning models.
- Create unit and integration tests for the methods that you have implemented using pytest (sample test methods to demonstrate your skills are sufficient).
- Implement automated documentation for written scripts and methods using Sphinx (sample documentations for selected implemented methods in Sphinx to demonstrate your skills are sufficient).
- Implement data pipelines using Airflow or Metaflow or Dagster (mandatory, choose either one framework to implement automated data pipelines).
- Implement model and data versioning with tracking using MLFlow or Kedro or Comet ML (mandatory, choose either one framework or any other framework to implement model versioning with tracking).
- Present results in the form of dashboard (Power BI or Grafana).

Submission deadline: **Three weeks** after receiving the case study.

Please submit your case study results directly via email to mei-chin.pang@basf.com.

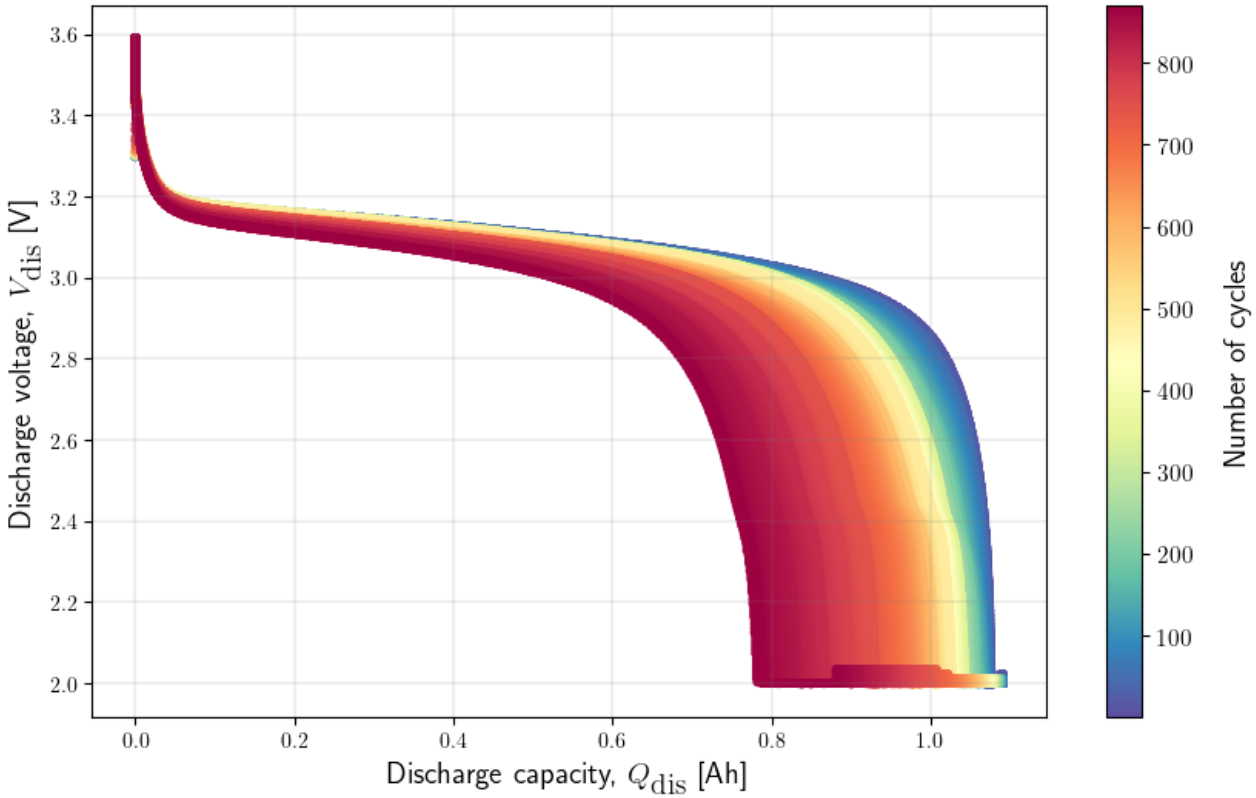
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Important submission remarks

- We are looking for clean and reproducible code with clear reasoning in both code comments and Sphinx documentation, for example:
  - Why do you choose this statistical method over the other methods?
  - How do you use this machine-learning model to improve the data quality? Why do you choose this model?
  - How do you implement automated data pipelines with model versioning?
- Please do not remove the anomalies manually. Here, we are interested in how you leverage statistics and machine-learning techniques to detect and remove the anomalies.
- Please follow PEP8 (Style Guide for Python Code) whenever possible in your code.
- Bonus: You demonstrate your cloud computing skills in Azure. As a student, you could apply for Azure cloud subscription for free.
- Please compile all results into a single zipped folder with a *README.md* on how to run your code:
  - Please compile all the libraries you have used in this case study in a *requirements.txt* file with the specific used versions.
  - Both submissions in the form of Python notebooks or Python scripts are acceptable.
  - If the files size are too big to be attached into an email, please create a separate submission share-folder (folder name: YOURNAME_data_science_case_study2025) using DropBox or Box or OneDrive and share the link with mei-chin.pang@basf.com. Please do not use the original share-drive to submit your solutions, this share-drive from BASF is opened to all candidates.

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Example plot without anomalies



While some of you may not have battery domain knowledge, here is a sample plot on a normal battery cell cycling dataset without anomalies:

- X-axis: Cell discharge capacity in the unit Ah
- Y-axis: Cell discharge voltage in the unit V.

The cell discharge profile is plotted over multiple cycles, indicated by the color bar. Darker red curves show how the cell performance has degraded over time with increasing cycles.

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